Turbocharger Assembly

Service and Diagnostic Manual

TITAN MOTORS MX-13 Engine Platform

Part Number: TC-5000

Revision: C

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Classification: Dealer Restricted

Supersedes: All previous revisions

Section 1: General Description

The TC-5000 Turbocharger Assembly is a variable geometry turbocharger designed for the TITAN MOTORS MX-13 engine platform used in Atlas T680, T880, W990, and Titan 579, 567 model families. The assembly provides forced induction across the full engine operating range and is integral to meeting current EPA emissions requirements through precise management of exhaust energy and air-fuel ratio control.

This manual covers the complete service lifecycle of the TC-5000 assembly including installation, theory of operation, diagnostic procedures, and preventive maintenance guidance. Technicians should be familiar with OEM diagnostic diagnostic software operation and TITAN MOTORS MX-13 engine fundamentals before performing work on the turbocharger system.

Section 2: Component Identification

The TC-5000 consists of seven major sub-component groups illustrated in Figure 2-1 below. Each component is identified by item number and described in Table 2-1.

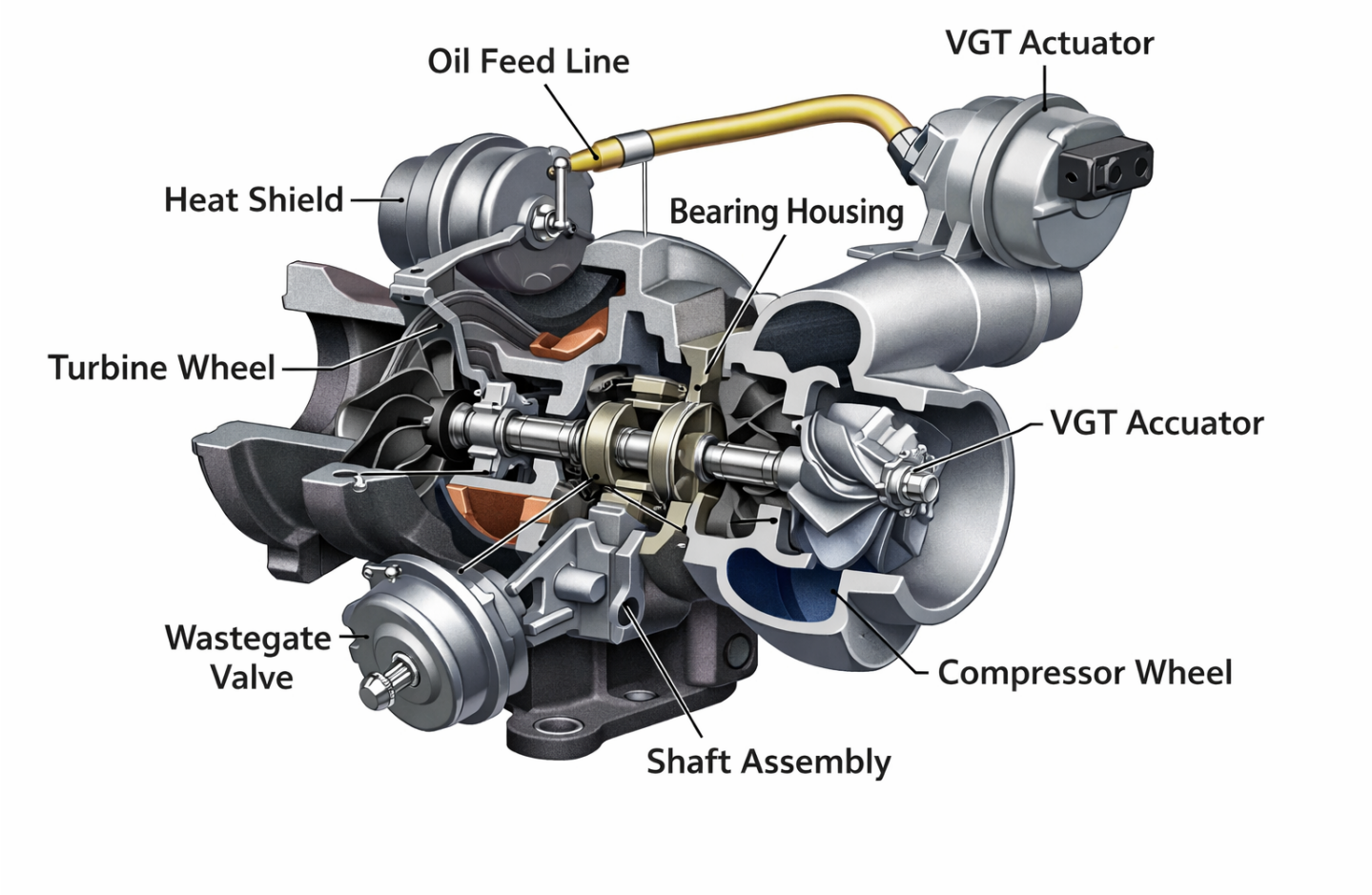

Figure 2-1: TC-5000 Turbocharger Assembly - Exploded View

Table 2-1: Sub-Component Identification

Item	Sub-Component	Description
1	Turbine Wheel	Inconel 713C investment casting. Converts exhaust gas thermal energy into rotational force. Rated to 128,000 RPM and 1,250 deg F continuous duty.
2	Compressor Wheel	Machined 2618-T61 aluminum alloy. Draws ambient air through the intake and delivers compressed charge air at 18-42 PSI boost to the intercooler circuit.
3	Bearing Housing	Central structural member containing the full-floating hydrodynamic journal bearing, oil feed and drain passages, and piston ring shaft seals.
4	VGT Actuator	Electronically controlled stepper-motor actuator. Modulates the turbine vane ring position to control exhaust gas velocity across the turbine wheel. Response time < 120 ms.
5	Wastegate Valve	Spring-loaded bypass valve. Diverts exhaust gas around the turbine when boost pressure exceeds 48 +/- 2 PSI crack pressure to prevent over-boost.
6	Oil Feed Line	Braided stainless steel line with restrictor orifice. Delivers filtered engine oil to the bearing housing at 22-65 PSI depending on engine speed.
7	Heat Shield	Stamped stainless steel thermal barrier. Isolates turbine housing radiant heat from adjacent wiring harnesses, hoses, and engine-bay components.

Section 3: Specifications

The following specifications apply to all TC-5000 variants unless noted otherwise.

Maximum Turbine Speed	128,000 RPM
Boost Pressure Range (sea level)	18 - 42 PSI
VGT Actuator Response Time	< 120 ms commanded to achieved
Oil Feed Pressure (at idle)	22 - 35 PSI
Oil Feed Pressure (at rated)	45 - 65 PSI
Exhaust Gas Temperature (max continuous)	1,250 deg F
Turbine Housing Material	High-silicon ductile iron
Bearing Type	Full-floating journal bearing
Wastegate Crack Pressure	48 +/- 2 PSI
Assembly Weight (dry)	47.5 lbs
Mounting Bolt Torque (V-band clamp)	11.5 ft-lbs
Oil Feed Line Fitting Torque	18 ft-lbs
Oil Drain Line Fitting Torque	22 ft-lbs
Coolant Supply Line (if equipped)	15 ft-lbs

Section 4: Installation and Removal

Removal Procedure

Prior to removal, allow the engine to cool for a minimum of 45 minutes. The turbine housing surface temperature can exceed 800 degrees F during operation and will retain significant heat for an extended period. Disconnect the battery ground cables before beginning work.

Begin by removing the charge air cooler pipe from the compressor outlet. Loosen the V-band clamp using a 10mm socket and carefully separate the joint. Cap the compressor outlet to prevent foreign object entry. Next, disconnect the oil feed line at the bearing housing fitting using a 14mm line wrench. Allow residual oil to drain into a suitable container. Disconnect the oil drain line at the lower bearing housing flange. Remove the four M8 bolts securing the oil drain flange and lower the drain tube clear of the assembly.

Disconnect the VGT actuator electrical connector by depressing the locking tab and pulling straight back. Do not pry on the connector body. Disconnect the wastegate actuator vacuum line by pressing the quick-connect release collar.

Remove the exhaust inlet V-band clamp at the turbine housing using a 12mm socket. The turbocharger can now be lifted free from the exhaust manifold studs. The assembly weighs approximately 47.5 pounds and requires two-hand support during removal.

Installation Procedure

Inspect the exhaust manifold flange surface for warpage using a straight edge and feeler gauge set. Maximum allowable deviation is 0.006 inches. Install a new turbine inlet gasket. Pre-lubricate the bearing housing with 30ml of clean 15W-40 engine oil through the oil feed port before mounting the assembly.

Position the turbocharger on the exhaust manifold studs and hand-tighten the V-band clamp. Torque the V-band clamp to 11.5 ft-lbs. Reconnect the oil feed line (18 ft-lbs), oil drain line (22 ft-lbs), VGT actuator connector, wastegate vacuum line, charge air cooler pipe, and inlet hose.

Section 5: Theory of Operation

Variable Geometry Turbocharging

The TC-5000 employs a variable geometry turbine to provide optimal boost pressure across the entire engine operating range. At low engine speeds, the ECM commands the VGT actuator to close the vane ring, reducing the effective nozzle area and increasing exhaust gas velocity across the turbine wheel. As engine speed and exhaust flow increase, the actuator progressively opens the vane ring to prevent over-boosting.

Bearing System Dynamics

The bearing housing supports the turbocharger shaft at rotational speeds exceeding 120,000 RPM through a full-floating hydrodynamic journal bearing. This bearing type relies entirely on a pressurized oil film to prevent metal-to-metal contact. The oil film thickness at rated speed is approximately 0.0015 inches.

Wastegate Function

The wastegate valve provides mechanical overpressure protection by diverting exhaust gas flow around the turbine wheel when boost pressure exceeds the crack pressure setting. The valve is spring-loaded to the closed position and opens progressively as boost pressure acts on the actuator diaphragm.

Section 6: Troubleshooting and Diagnostics

This section describes common failure symptoms, their probable root causes, and the recommended diagnostic approach. Technicians should use this information in conjunction with OEM diagnostic diagnostic software to isolate faults.

6.1 Gradual Loss of Boost Pressure

Operators may report a gradual decline in engine pulling power accompanied by dark gray or black exhaust smoke under load. This condition often develops slowly over thousands of miles, making it difficult to identify when it began. The most common cause of gradual boost loss is degradation of the wastegate valve seat. Over extended service, the wastegate valve seat erodes from the high-velocity, high-temperature exhaust gas that contacts the sealing surface each time the valve cycles. As the seat erodes, the wastegate fails to fully seal at pressures below its crack point, allowing exhaust energy to bypass the turbine wheel continuously. Drivers in flat terrain may not notice the reduced boost, but operators who regularly encounter mountain grades or operate at altitude will report that the truck lacks the pulling power it once had, particularly above 6,000 feet elevation.

If wastegate seat erosion is ruled out, attention should turn to the VGT actuator. Carbon buildup on the internal vane ring can increase actuator resistance, preventing the vane ring from closing fully at low engine speeds. Perform a VGT actuator sweep test through OEM diagnostic. The actuator should move smoothly from 0 to 100 percent position with no hesitation. Monitor current draw during the sweep: normal range is 0.4 to 1.8 amps. Current spikes above 2.5 amps indicate excessive mechanical resistance from carbon deposits on the vane ring or degradation of the actuator internal gear mechanism.

6.2 Abnormal Turbocharger Noise

A high-pitched continuous metallic whine from the turbocharger center section, particularly noticeable at idle and low-speed operation, suggests bearing distress in the bearing housing. This sound occurs when the oil film supporting the shaft journal has been compromised, allowing intermittent metal-to-metal contact. The most frequent cause of oil film disruption is restricted flow through the oil feed line. Over extended service life, the oil feed line develops internal varnish buildup that reduces oil flow to the bearing housing below the minimum required to maintain a full hydrodynamic film. The whine may initially appear only during cold starts when oil viscosity is highest, leading technicians to dismiss it as a temporary condition. If the oil feed line is not replaced within approximately 8,000 to 10,000 miles of initial symptom appearance, the bearing sleeve will develop scoring that requires replacement of the entire bearing housing assembly.

A warbling or oscillating tone that varies independent of engine load indicates that the turbine wheel balance has been disturbed. This can result from foreign object damage to the turbine blades, erosion from prolonged exposure to elevated exhaust gas temperatures, or from the turbine wheel contacting the housing due to excessive shaft play from bearing wear. Measure shaft end-play using a dial indicator through the oil drain port. Maximum allowable end-play is 0.003 inches.

6.3 Oil in Charge Air System

The presence of engine oil pooling or dripping at the charge air cooler connections downstream of the compressor indicates a compressor-side seal failure in the bearing housing. The compressor wheel piston ring seal prevents engine oil that lubricates the bearing from migrating along the shaft into the compressor housing. Seal failure is often secondary to bearing wear that allows excessive shaft radial movement, distorting the seal contact surface. Inspect the compressor wheel for rub marks on blade tips which would confirm bearing-related shaft displacement.

6.4 Intermittent Power Lag During Acceleration

A subtle increase in turbo lag during acceleration events, most noticeable when the driver tips in the throttle from idle or low cruise RPM, is the earliest indicator of VGT actuator degradation. The actuator develops increased internal resistance typically due to carbon infiltration past the housing seals or wear in the internal gear mechanism. At this stage the ECM may not log a fault code because the actuator still achieves commanded position - it simply takes longer to get there. The operator may describe it as the truck feeling sluggish on some days but not others. If not addressed within approximately 15,000 miles, the degradation forces the ECM to compensate by modifying fueling strategy, increasing exhaust gas temperatures by 50 to 100 degrees F above normal range, which accelerates thermal fatigue in the turbine wheel at the blade root fillets.

6.5 Heat Shield Failure Effects

A cracked, warped, or loose heat shield allows radiant thermal energy from the turbine housing to reach adjacent engine bay components that are not designed for high-temperature exposure. The most common secondary damage is degradation of wiring harness insulation and softening of rubber coolant hoses in the immediate vicinity. If a vehicle presents with unexplained sensor faults or coolant leaks near the top of the engine between the manifold and valve cover, inspect the heat shield condition before beginning electrical or cooling system diagnosis.

Section 7: Preventive Maintenance

At each B-service interval (50,000 miles), visually inspect all external connections and mounting hardware. Check V-band clamp torque at the turbine inlet and compressor outlet. Inspect the VGT actuator electrical connector for heat damage or corrosion. Verify that the heat shield is securely mounted and has not developed cracks or warpage.

At each C-service interval (100,000 miles), perform a VGT actuator sweep test using OEM diagnostic and record actuator response time and current draw. Compare to previous service records to identify trending degradation. Inspect the oil feed line external surface for cracking, chafing, or deformation.

At 200,000 miles, regardless of symptom status, TITAN MOTORS recommends replacement of the oil feed line assembly as a preventive measure. Field data shows that oil feed lines in service beyond 200,000 miles have a significantly elevated risk of internal varnish restriction.